

A *bug* is a problem in the way a program runs. *Debugging* is the process of identifying the cause of a bug and modifying code to fix the issue.

One of the first steps in debugging is to understand error messages. They tell you where the interpreter had a problem and what kind of problem occurred.

Logical errors don't result in error messages, so you need to look at the program execution to resolve them. To debug logical errors, examine the value of variables at different points in the program's execution. Do this by inserting `print()` calls into the code at strategic points, logging the values of key variables during execution, or using the debugging tools available in most IDEs. Also, write tests for your code to see which parts are working and which are not.